



SOMAIYA
VIDYAVIHAR UNIVERSITY

08.05.2024(E)

Semester: Jan 2024-April 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration: 3 Hrs.
Programme code: 03		
Programme: B. Tech Electronics and Telecommunication Engineering	Class: SY	Semester: IV (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC
Course Code: 116U03C401	Name of the Course: Mathematics For Communication Engineering-II	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks														
Q1	Solve any Four of the following.	20														
i)	Determine whether the vectors $v_1 = (1, -1, 1)$, $v_2 = (0, 1, 2)$, $v_3 = (3, 0, -1)$ form the basis for R^3	5														
ii)	Evaluate $\int_C ydx + x^2dy$ along $y = x$ from $(0, 0)$ to $(2, 2)$	5														
iii)	Show that the function $f(z) = z^2$ is analytic and find its derivative.	5														
iv)	Calculate the coefficient of correlation from the data: $N = 10, \sum x = 100, \sum y = 150, \sum (x - 10)^2 = 180, \sum (y - 15)^2 = 215, \sum (x - 10)(y - 15) = 60$ Where x, y denote the actual values	5														
v)	If the probability that an individual suffers a bad reaction from a particular injection is 0.001, determine the probability that out of 2,000 individuals more than two individuals will suffer a bad reaction using Poisson distribution. (Given : $e^{-2} = 0.1353$).	5														
vi)	Covert the following LPP to standard form Minimize $z = -3x_1 + 2x_2 - x_3$ Subject to $x_1 - 3x_2 + 2x_3 \geq -6, 3x_1 + 4x_3 \leq 3, -3x_1 + 5x_2 \leq 4, x_1, x_2, x_3 \geq 0$	5														
Q2 A	Solve the following.	10														
i)	Find optimum of the function $z = x_1x_2 + 9x_1 + 6x_3 - x_1^2 - x_2^2 - x_3^2$	5														
ii)	Evaluate $\int_C \frac{e^{2z}}{(z-1)(z-2)} dz$, where C is the circle $ z = 3$	5														
OR																
Q2 A	Five dice are thrown together 96 times. The number of times 4, 5 or 6 was obtained is given below. <table border="1"><tr><td>No of times 4,5 or 6 was obtained</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Frequency</td><td>1</td><td>10</td><td>24</td><td>35</td><td>18</td><td>8</td></tr></table> Fit a Binomial distribution if the nature of the dice is not known.	No of times 4,5 or 6 was obtained	0	1	2	3	4	5	Frequency	1	10	24	35	18	8	10
No of times 4,5 or 6 was obtained	0	1	2	3	4	5										
Frequency	1	10	24	35	18	8										

Q 2 B	Solve any One of the following.	10
i)	Let the vector space P_2 have the inner product defined by $\langle p, q \rangle = \int_{-1}^1 p(x) \cdot q(x) dx$. Use Gram-Schmidt process to transform the standard basis $S = \{1, x, x^2\}$ into an orthonormal basis	10
ii)	Using Big M method to solve the LPP Minimize $z = x_1 + x_2$ Subject to $2x_1 + x_2 \geq 4$, $x_1 + 7x_2 \geq 7$, $x_1, x_2 \geq 0$	10
Q3	Solve any Two of the following.	20
i)	Prove that the set $S = \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} : a, b \in \mathbb{R} \right\}$ form a vector space with usual addition of matrices and scalar multiplication	10
ii)	Expand $f(z) = \frac{1}{(z-1)(z-2)}$ for the regions (a) $ z < 1$ (b) $1 < z < 2$ (c) $ z > 2$	10
iii)	Monthly salaries of 1000 workers have a normal distribution with mean of Rs. 575 and a standard deviation of Rs. 75. Find the numbers of workers having monthly salaries between Rs. 500 and Rs. 625. Also find the minimum salary of the highest paid 200 workers.	10
Q4	Solve any Two of the following.	20
i)	Using Cauchy's residue theorem, evaluate $\int_{-\infty}^{\infty} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx$	10
ii)	The regression lines of a sample are $x + 6y = 6$, and $3x + 2y = 10$. Find (a) Sample means $\bar{x}$ and $\bar{y}$, (b) coefficient of correlation between x and y . Also estimate y when $x = 12$.	10
iii)	Solve the LPP by Simplex method Maximize $z = 10x_1 + x_2 + x_3$ Subject to $x_1 + x_2 - 3x_3 \leq 10$, $4x_1 + x_2 + x_3 \leq 20$, $x_1, x_2, x_3 \geq 0$	10
Q5	Solve any Four of the following.	20
i)	If the probability of a rivet being defective is 0.1, find the probability that out of 5 randomly chosen rivets none will be defective.	5
ii)	If X is a normal variate with mean 30 and standard deviation 5. Write SNV and find $P(26 \leq X \leq 40)$	5
iii)	In a Poisson distribution if $P(X = 3) = \frac{2}{3} P(X = 4)$. Find the mean and the Standard deviation of the distribution.	5
iv)	Find p if $f(z) = r^2 \cos 2\theta + ir^2 \sin p\theta$ is analytic.	5
v)	Find the residues of the function $f(z) = \frac{e^z}{z^2 + a^2}$ at its poles	5
vi)	For NLPP, Optimize $z = 4x_1 + 8x_2 - x_1^2 - x_2^2$ Subject to $x_1 + x_2 = 4$, $x_1, x_2 \geq 0$ Construct a Lagrangian function and obtain stationary point.	5



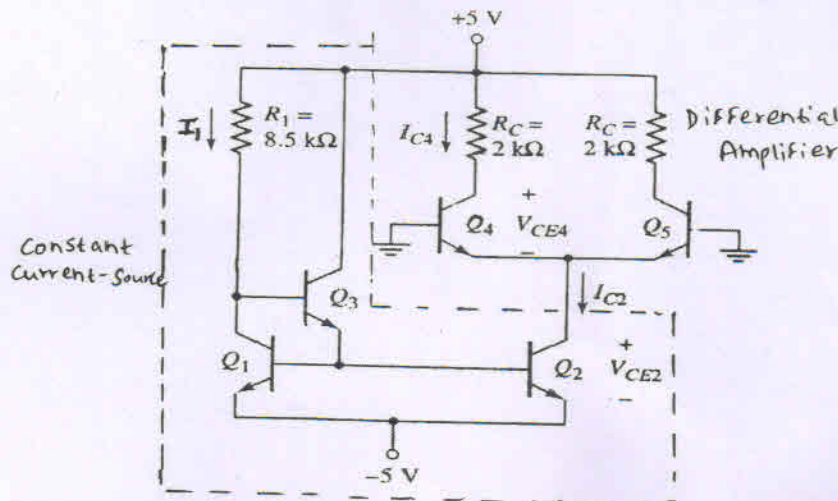
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10.05.2024(E)

Semester: January 2024 –April 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03	Class: SY	Semester: IV (SVU 2020)
Programme: B.TECH.		
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: EXTC	
Course Code: 116U03C402	Name of the Course: Analog Electronics	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What is the need of multi-stage amplifier?	5
ii)	Draw a basic two-transistor constant current source. State the relation between output current and reference current.	5
iii)	State the features of an ideal op-amp.	5
iv)	Why heat sink is necessary in case of power amplifier?	5
v)	Draw the DC load line along with collector waveforms showing Q-point each for Class A, Class B, Class AB and Class C power amplifiers.	5
vi)	State and explain the Barkhausen Criteria for sustained oscillations.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	How cross-over distortion is avoided in class AB amplifier?	5
ii)	Why power amplifier is called as a large signal amplifier?	5
	OR	
Q2 A	Describe the working of transformer coupled Class A Power amplifier. Derive the expression for maximum conversion efficiency.	10
Q 2 B	Solve any One	10
i)	Explain the circuit given below. The transistor parameters are: $\beta=100$, $V_{BE(on)}=0.7V$, and $V_A=\infty$. Determine I_1 .	10



ii)	Derive an expression for voltage gain of inverting amplifier using Op-Amp. Explain how a basic inverting amplifier configuration of a Op-Amp can be used as following circuits: (i) Sign changer (ii) Scale changer	10
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Draw the High frequency equivalent circuit of Common source MOSFET amplifier circuit. For the Common source NMOSFET amplifier circuit; Given that : the parameters are , $R_1 = 500 \text{ K}$, $R_2 = 225 \text{ K}$, $R_S = 1 \text{ K}$, $R_D = 5 \text{ K}$, $R_L = 10 \text{ K}$, $R_{Si} = 1 \text{ K}$, $g_m = 10 \text{ mS}$, $k_n = 1 \text{ mA/V}^2$, $V_{TN} = 2 \text{ V}$, $\lambda = 0$, $C_{gs} = 5 \text{ pF}$, $C_{gd} = 1 \text{ pF}$. Determine $A_v(\text{mid})$, 3-dB F_H .	10
ii)	A high frequency BJT is biased at $I_{CQ} = 0.5 \text{ mA}$ and has parameters $C_\mu = 0.15 \text{ pF}$, cut-off frequency at unity gain (f_T) = 5 GHz, and $\beta_0 = 150$. Determine C_π and bandwidth f_β .	10
iii)	Draw circuit diagram of CE-CB cascode amplifier. Deduce the expressions of A_v , Z_i and Z_o for the same using hybrid-pi model.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Draw and explain the four basic negative feedback configurations of Op-Amp.	10
ii)	Design a Circuit using OP-AMP to provide the output voltage V_o - $V_o = -4 V_1 - 2V_2 - V_3$	10
iii)	Design a first order high pass active filter using OP-AMP, with cut-off frequency of 3 KHz and voltage gain of 10.	10

Que. No.	Question	Max. Marks
Q5	Write notes on any four	20
i)	Darlington pair amplifier	5
ii)	Compare Wilson and Widlar constant current source	5
iii)	Schmitt trigger using Op-Amp	5
iv)	RC phase shift oscillator	5
v)	Miller effect	5
vi)	Hartley Oscillator	5

13/5/2024(E)


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Maximum Marks: 100		Semester: January 2024 –April 2024		Examination: ESE Examination		Duration:3 Hrs.	
Programme code: 03				Class: SY		Semester: IV(SVU 2020)	
Programme: UG, Electronics and Telecommunication Engineering							
Name of the Constituent College: K. J. Somaiya College of Engineering				Name of the department: EXTC			
Course Code: 116U03L403		Name of the Course: Communication Systems					
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary							

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	A DSBFC AM is using 800kHz carrier with amplitude of 40V and modulating signal with 25kHz with modulation index 0.4 and carrier power of 400W. Determine : i) USB and LSB power ii) Total sideband power iii) Total transmitted power iv) Draw power spectrum.	5
ii)	The signal power and noise power measured at the input of an amplifier are 150 μ W and 1.5 μ W respectively. If the signal power at the output is 1.5W and noise power is 40 mW, calculate the amplifier noise, factor, noise figure and equivalent noise temperature if T_0 is 17°C.	5
iii)	Compare between DSBFC, DSBSC and SSB on the basis of Power requirement, Bandwidth, Advantages and Disadvantages.	5
iv)	When a super heterodyne radio receiver is tuned to 555kHz and its local oscillator frequency is 1010kHz. Calculate image frequency. If f_s is changed to 655kHz what will be I.F. generated?	5
v)	How FM can be obtained from PM?	5
vi)	Explain PAM generation and draw PAM circuit input and output waveforms	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain AM diode detector circuit with waveforms.	5
ii)	What is need of Pre-emphasis and De-emphasis networks? Explain Pre-emphasis circuit with diagrams.	5
	OR	
Q2 A	Explain Armstrong method of FM generation with block diagram and necessary phasor diagrams.	10

Q 2 B	Solve any One	10
i)	For an FM modulator with a peak frequency deviation $\Delta f = 10\text{kHz}$, a modulating signal frequency $f_m = 10\text{kHz}$, $V_c = 10\text{V}$ and $f_c = 500\text{kHz}$, if following are Bessel function values: $J_0 = 0.77$, $J_1 = 0.44$, $J_2 = 0.11$, $J_3 = 0.02$ Determine: i) modulation index ii) Minimum bandwidth required using Carson's rule iii) How many sideband pairs iv) Draw the frequency spectrum	10
ii)	One input to a conventional AM modulator is a 500kHz carrier with amplitude of 20Vp . The second input is 10kHz modulating signal that is of sufficient amplitude to cause a change in the output wave of $\pm 7.5\text{Vp}$. Determine i) Upper and lower side frequencies. ii) Modulation coefficient and % modulation iii) Peak amplitude of the modulated carrier and the upper and lower side frequency voltages iv) Max and Min amplitude of the envelope v) Equation for the modulated wave vi) Draw the o/p spectrum vii) Sketch the output envelope	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain block diagram of Super heterodyne radio receiver for DSBFC system and draw necessary waveforms at the input and output of the system.	10
ii)	Explain different types of tracking methods used in radio receiver.	10
iii)	Explain the concept of I.F and image frequency with a suitable example.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	What is sampling theorem and Nyquist rate? Explain with a simple example and waveforms.	10
ii)	Compare between PAM, PWM and PPM.	10
iii)	Explain Adaptive delta modulation with neat diagram and waveforms.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Write a short note on different distortions in diode detector based on RC time constant.	5
ii)	What are advantages of FM over AM?	5

iii)	Explain simple AGC circuit with its advantages and disadvantages.	5
iv)	How noise is characterized by different methods?	5
v)	Explain flat top PAM with neat diagrams.	5
vi)	Explain Frequency division multiplexing used in communication systems.	5

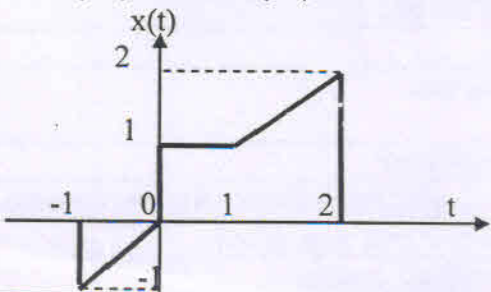


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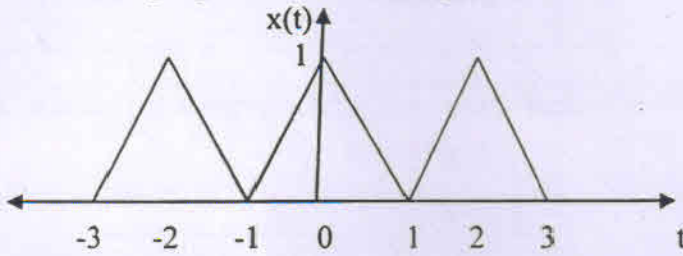
15/5/2024 (E)

Semester: January 2024 –April 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03	Class: SY	Semester: _IV_(SVU 2020)
Programme: BTech EXTC	BTech EXTC	
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC
Course Code: 116U03C404	Name of the Course: Signals and Systems	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Classify following system as static / dynamic ,causal / anticausal , linear / non linear. Justify your answer. $y[n] = x[n] + n x[n-1]$	5
ii)	Find Z transform and ROC for signal $x[n] = (0.5)^n u[n]$	5
iii)	Define Discrete time fourier transform (DTFT) with necessary equation. Find DTFT for $x[n] = [1 -2 3 -4]$	5
iv)	Sketch following signal $X(t) = 1; -1 < t < 1$ $= 2; 1 < t < 2$ $= 1; 2 < t < 4$	5
v)	Compare deterministic and random signals with suitable examples	5
vi)	Find inverse Z transform if $X(Z) = 1 / (1 + 4.5 Z^{-1} + 3.5 Z^{-2})$ If ROC $ Z > 3.5$.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Determine whether the signal is periodic. If periodic find the fundamental period. $x(n) = \cos\left(\frac{5\pi}{7}n\right) + 2 \sin\left(\frac{3\pi}{8}n\right)$	5
ii)	Sketch $x(t-2)$ and $2.x(t/2)$ for the following signal 	5
OR		
Q2 A	Determine whether following signals are Energy or power signal. Calculate their energy or power. i) $x(t) = u(t) - u(t-3)$ ii) $x[n] = 3 (0.5)^n u[n]$	10

Q 2 B	Solve any One	10
i)	Perform convolution of following D.T. signals by any method. $x[n] = 1$ for $0 \leq n \leq 2$ $= 0$ otherwise $h[n] = 1$ for $0 \leq n \leq 2$ $= -1$ for $3 \leq n \leq 5$ $= 0$ otherwise	10
ii)	If a LTI system with impulse response $h(t)=u(t)$ has input signal $x(t)=e^{-2t}u(t)$ find the output response of the system using convolution integral.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Express following signal in terms of Exponential Fourier series.  $x(t) = 1 + t$; for $-1 \leq t \leq 0$ $= 1 - t$; for $0 \leq t \leq 1$	10
ii)	State and prove Convolution property of Fourier transform	10
iii)	Find C.T. Fourier Transform for a signum function	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Draw pole zero plot for the transfer function of C.T. system given below. Find inverse Laplace Transform for all possible ROCs. $H(S) = \frac{s^2 + s - 2}{s(s+2)(s-3)}$	10
ii)	A DT system is described by $y(n)-2y(n-1)-3y(n-2)=x(n-1)$ i) Determine transfer function of the system ii) Draw pole zero plot and comment on stability	10
iii)	Determine step response of an LTI system whose impulse response $h[n]$ is given by $a^{-n} u[-n]$. Use Z transform method.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Relationship between Z transform and Laplace transform. (Any two points)	5
ii)	Sketch the signal $x(t) = -u(t-1)$ where $u(t)$ is unit step signal.	5
iii)	Whether following statement is true or false? Justify. All static systems are causal systems	5
iv)	What is the condition of periodicity of any D.T. signal? Give an example	5
v)	Find C. T. Fourier transform of unit step function $u(t)$	5
vi)	If $H(Z) = 1 + z^{-1} + 3z^{-2} + 4z^{-3}$ Find its inverse Z transform	5



Semester: January 2024 –April 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03	Class: 5 ^Y B.Tech	Semester:IV th (SVU 2020)
Programme: Electronics and Telecommunication Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: EXTC	
Course Code: 116U03C405	Name of the Course: Electromagnetic Field Theory	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Point charges 1mC and -2mC are located at (3,2,-1)m and (-1,-1,4)m respectively. Calculate the electric force on 10nC charge located at (0,3,1)m.	5
ii)	Explain Laplace's and Poisson's equations and express them for Cartesian coordinate system.	5
iii)	A long straight wire carries a current I= 1Amp. At what distance is the magnetic field H= 1A/m will be obtained?	5
iv)	Justify $B_{n1}=B_{n2}$ i.e. normal component of B is continuous across the surface for magnetic boundary condition.	5
v)	Write the set of Maxwell's equations in integral and point form.	5
vi)	A uniform wave in air has $E=10\cos(2\pi \times 10^9 t - \beta z)\mathbf{a}_y$ Calculate: i) Frequency f in Hz ii) Frequency ω in rad/s ii) Velocity v iii) Phase constant β iv) Wavelength λ	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Plane $z = -10$ m carries charge 10 nC/m^2 . What will be the electric field intensity in free space at the origin?	5
ii)	Given the potential $V=(2x^2y - 5z)$ and a point P(-4, 3,6) m. Find V, E and D at P.	5
	OR	
Q2 A	Find the work done in moving the point charge of $3\mu\text{C}$ from Origin to P(2,-1,4)m through the field $E= (2xyz \mathbf{a}_x + x^2z \mathbf{a}_y + x^2y \mathbf{a}_z)\text{V/m}$ through the path (0,0,0)m to (2,0,0)m to (2,-1,0)m to (2,-1,4)m	10
Q2 B	Solve any One	10
i)	Determine the expression for magnetic field intensity due to an infinite sheet carrying current.	10

ii)	Given potential field $V = 3xy^2z + K yz^3$ V/m. Find K if potential field satisfies Laplace's equation, also find E at (2,3,1).	10
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Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	A current distribution gives rise to the vector magnetic potential $A = (x^2y) \mathbf{a}_x + (y^2x) \mathbf{a}_y - (4xyz) \mathbf{a}_z$ Wb/m. Calculate the following: The flux through the surface defined by $z=4$, $0 \leq x \leq 2$, $-1 \leq y \leq 2$	10
ii)	A current filament of 10 A in $\mathbf{a}_y$ direction is parallel to y axis at $x=2$ m and $z = -2$ m. Find H at the origin.	10
iii)	In a certain conducting region $H = yz(x^2 + y^2) \mathbf{a}_x - y^2 xz \mathbf{a}_y + 4x^2 y^2 \mathbf{a}_z$ (A/m) i) Determine J at (5,2,-3)m ii) Show that $\nabla \cdot B = 0$	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	A 10GHz plane wave travelling in free space in z direction has an amplitude of $E_x = 10$ V/m. Find velocity, wavelength, phase constant, intrinsic impedance and amplitude and direction of H .	10
ii)	Calculate the skin depth for a medium with conductivity 100 mho/m, for: i) relative permeability 2 ii) relative permeability 3 at i) 50Hz ii) 1MHz	10
iii)	A plane wave of 200MHz travelling in free space impinges normally on a large block of material having $\epsilon_r = 4$, $\mu_r = 9$ and $\sigma = 0$. Determine η_1 , η_2 , β_1 , β_2	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Explain any one real time application based on electromagnetic waves.	5
ii)	Explain Biot- Savart's law.	5
iii)	Write short note on Scalar and Vector magnetic potential.	5
iv)	Write a short note on Poynting theorem.	5
v)	Explain Continuity equation for current.	5
vi)	Effects of electromagnetic radiations on human body.	5